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Published: 29 December 2025

Citation in BibTeX format

UbiComp '25:The 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing / ISWC ACM International Symposium on Wearable Computers
October 12 - 16, 2025
Espoo, Finland

Conference Sponsors:
SIGMOBILE
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UbiComp Companion '25: Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (October 2025)
https://doi.org/10.1145/3714394.3756203
ISBN: 9798400714771

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Abstract

This summary paper provides an overview of Task 2 from the Sussex-Huawei Locomotion Challenge. In this task, the specific category of human activity recognition was intentionally undisclosed to participants. Under these conditions, teams were expected to develop classifiers—potentially utilizing foundation models—trained on publicly available datasets, with the objective of generalizing to an unseen test set provided by the organizers. The ranking criterion was the F1 score. This paper presents a comparative analysis of the methodologies employed by participants. Additionally, it reports on further experiments conducted by the organizers using the foundation models MOMENT and Chronos.

CCS Concepts

• **General and reference → Surveys and overviews**; • **Computing methodologies → Neural networks**.

Keywords

large-scale human activity sensing corpus, human activity recognition, foundation models

ACM Reference Format:

Tsuyoshi Okita, Kosuke Ukita, Asahi Miyazaki, Daichi Kubota, Jukichi Ota, Naoki Kagiyama, Asahi Nishikawa, Daichi Nagayasu, Syunya Tomitaka, Daisuke Nozaki, Yuki Odo, Raku Yamashita, Xiaolong Ye, Huayu Gao, Kazuki Okahashi, Koki Matsuishi, Masaharu Kagiyama, Kodai Hirata, Haruki Kai, Lin Wang, Hristijan Gjoreski, Mathias Ciliberto, Paula Lago, Kazuya Murao, and Daniel Roggen. 2025. Foundation Models to Tackle Activity Recognition in Unknown Domain: Sussex-Huawei Locomotion Challenge 2025 Task 2. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25), October 12–16, 2025, Espoo, Finland*. ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3714394.3756203

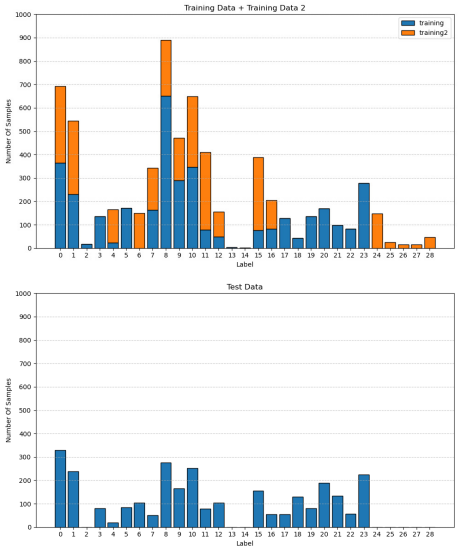
1 Task Description

In the Sussex-Huawei Locomotion Challenge [11] consisting of Tasks 1 and 2, Task 2 addresses the challenge of human activity recognition under cross-domain settings and unseen testing dataset, where models must generalize across different data distributions. Specifically, participants are required to classify activities from IMU data, e.g., acceleration, gyroscope, magnetometer, using training data that differ in distribution from the unseen test data. Participants may use any publicly available labeled or unlabeled datasets. The SHL 2025 dataset is suggested as a benchmark, and the Kyutech IMU dataset [7] is provided as an auxiliary unlabeled dataset. The test dataset with unknown domain to the participants is collected from a distinct domain.

Participants are encouraged to explore the use of foundation models trained via self-supervised learning, domain adaptation, and temporal modeling techniques. Pretrained models such as Chronos [2], MOMENT [5], and other time-series foundation models may be employed. Alternatively, large language models (LLMs), including ChatGPT, Gemini, and Claude, may also be utilized where appropriate. The flexibility of the task allows participants to leverage both domain-specific time-series models and general-purpose foundation models to develop innovative solutions.

Two of the three participating teams adopted the recently proposed MOMENT time-series foundation model, while the remaining team utilized a Transformer-based classifier. The teams leveraging MOMENT employed a range of innovative strategies to address domain shift, including extensive data augmentation during fine-tuning, semi-supervised learning frameworks, and complex multi-stage training pipelines. This summary highlights the key technical contributions of each team, compares their architectural choices and implementation strategies.

Figure 1: Distribution of training datasets 1 and 2 (the upper figure), and testing dataset (the lower figure).



2 Datasets and Software

For Task 2, three types of datasets were provided: (1) the Kyutech IMU Dataset [7], (2) the Sussex-Huawei Locomotion Challenge 2025 Dataset [4, 10], and (3) the SHL 2025 Task 2 testing dataset (a fine-tuning dataset), which includes Training Dataset 1, Training Dataset 2, and a Testing Dataset. (1) and (3) consist of the IMU sensor data[1]

(2) and (3) SHL are provided to support the training of both foundation models and Transformer-based models. (1) is reserved exclusively for target domain evaluation. Additionally, we provide a pretrained model, SENvT [6], which serves as a representative

[1]The IMU sensor data for the targeted testing dataset include: (1) Linear acceleration: (a_x, a_y, a_z), (2) Gravitational force ratio: (gF_x, gF_y, gF_z), (3) Gyroscope: $(\omega_x, \omega_y, \omega_z)$, and (4) Magnetometer: (B_x, B_y, B_z). Standard accelerometer readings can be computed by combining linear and gravitational components: $(ax + gFx,\ ay + gFy,\ az + gFz)$. This constituency is the same for fine-tuning training datasets of 1 and 2.

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UbiComp Companion '25, Espoo, Finland
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ACM ISBN 979-8-4007-1477-1/2025/10
https://doi.org/10.1145/3714394.3756203

Table 1: Comparison of Team Methodologies for Task 2.

Team	core model	train strategy	input features	domain shift strategy	key novelty	post-processing
TDU-BSA2[8]	MOMENT	fine-tuning	raw data	fine-tuning with online data augmentation	time-series-aware post-processing	Yes(Median Filter, LightGBM)
SIAT-BIT[9]	Transformer	-	handcrafted features	semi-supervised learning (MixMatch, Pseudo-labeling)	domain-aware MixMatch for zero-target-label scenario	
UT-IR[1]*	MOMENT-1-large	fine-tuning	raw data (15 channels)	advanced fine-tuning with multi-stage schedule	hybrid augmentation and learning	rate schedule cross-validation warm-start

Team	pred	test	cpu	gpu	train	test	lang	library	model size	win size	datasets (pretraining)	feature
TDU-BSA2	67.1 %	65.0 %	i9-12900k 16core @3.2GHz 64G	RTX 3080	1 hour	1 min	python	PyTorch	145 MB	5 sec		Acc Gyr Mag
SIAT-BIT	58.3 %	76.5 %	12core @3.7GHz 128GB	RTX 3090	2.5 hour	1 sec	python	Scikit-learn TSFEL	11.9 MB	5 sec	SHL2025 KyutechIMU PAMAP2 UCI-HAR mHealth RealWorld KU-HAR	Acc Gyr Mag
UT-IR	65.0 %	65.0 %	26core 6230R @2.10GHz 187GB	2× V100 32GB	17 hour	5 min	python	PyTorch momentfm lmdb scikit-learn	1.3 GB	5 sec	SHL2025	Acc Gyr Mag

Table 2: Classification labels of SHL 2025 Task 2 fine-tuning dataset of 29 class is shown.

label	action	label	action	label	action	label	action
badminton		gym		soccer		volleyball	
0	serve	7	benchpress	12	dribble	17	attack
1	smash	8	dumbbell	13	one-on-one defense	18	block
basketball		9	lat pull down	14	one-on-one offense	19	jump floater
2	dribble	10	running machine	15	pass	20	overhand pass
3	free throw	11	squat	16	shoot	21	serve
4	lay up	25	pull up	27	ball release	22	partner pass
5	pass	26	sit up	28	goal keeper	23	underhand pass
6	shoot						
24	quick pass						

foundation or Transformer model for benchmarking and further fine-tuning.

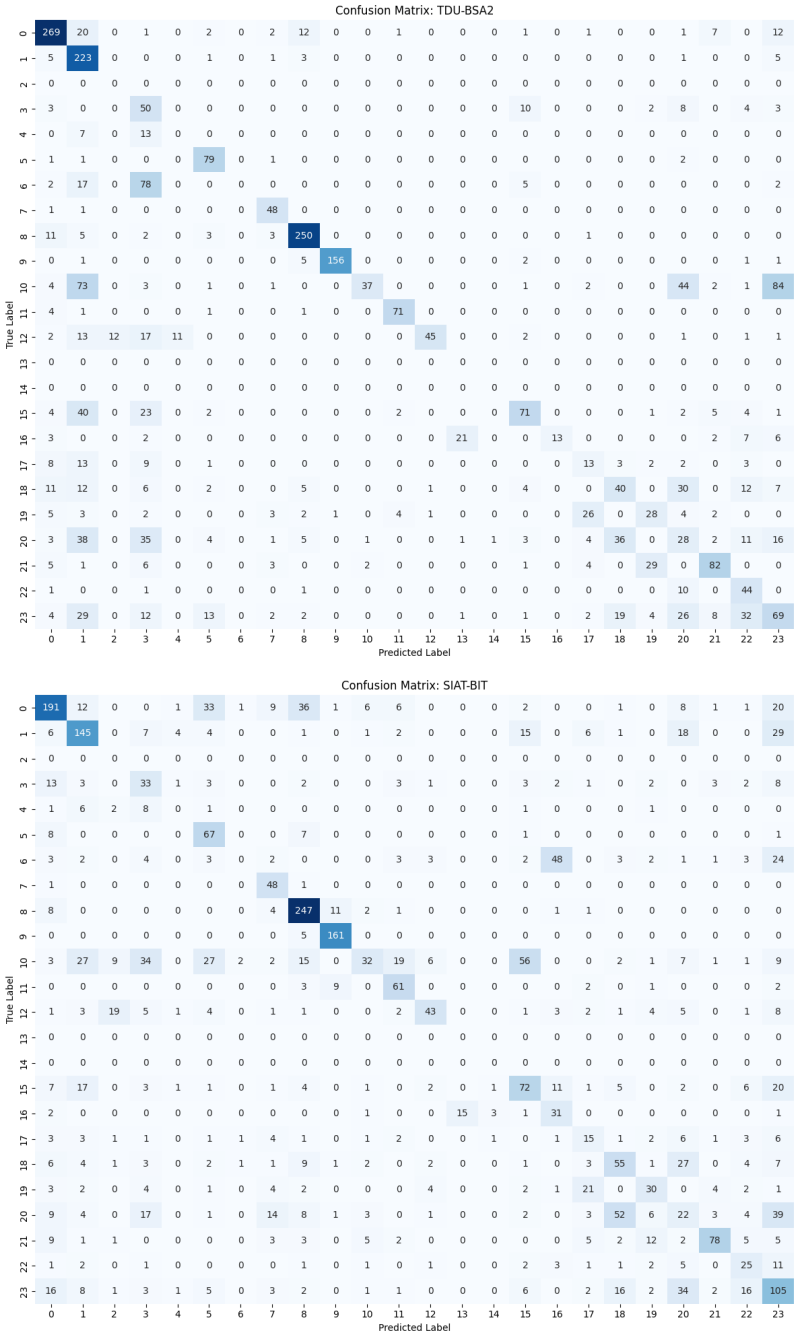
3 Methodologies of Participants

The methodologies employed by the three teams, TDU-BSA2 [8], SIAT-BIT [9], and UT-IR [1][2], are summarized in Table 1. While all teams adopted Transformer-based architectures, their approaches to model selection, feature engineering, and domain adaptation diverged significantly.

Notably, both TDU-BSA2 and UT-IR successfully fine-tuned the MOMENT foundation model, highlighting its effectiveness for cross-domain time-series classification tasks. Despite the convergence

[2]We allowed the participated team does not necessarily submit two papers both for Tasks 1 and 2. This is since we expect that some team may reuse most of the architecture for both tasks. Under this rule, UT-IR team submitted the paper for Task 1 [1] but not for Task 2. The overall architecture is different in Tasks 1 and 2, but it may be that the description of MOMENT is overlapped in the paper.

Figure 2: Confusion matrices of TDU-BSA2 and SIAT-BIT teams.



around Transformer-based architectures, the strategies for addressing the core challenge of domain shift varied considerably among the teams.

The **TDU-BSA2** team employed a standard fine-tuning approach augmented with a sophisticated, task-specific post-processing pipeline. This highlights the continued value of domain expertise, demonstrating that carefully crafted heuristics can provide a competitive edge even when using strong pretrained models. The **SIAT-BIT** team's semi-supervised learning framework is particularly noteworthy, as it is well-suited for scenarios where no labeled data from the target domain is available. Moreover, their integration of handcrafted features distinguishes their method from the end-to-end learning strategies adopted by the other teams. The **UT-IR** team exemplified the "scale is all you need" philosophy, leveraging

Figure 3: Confusion matrix of UT-IR team.

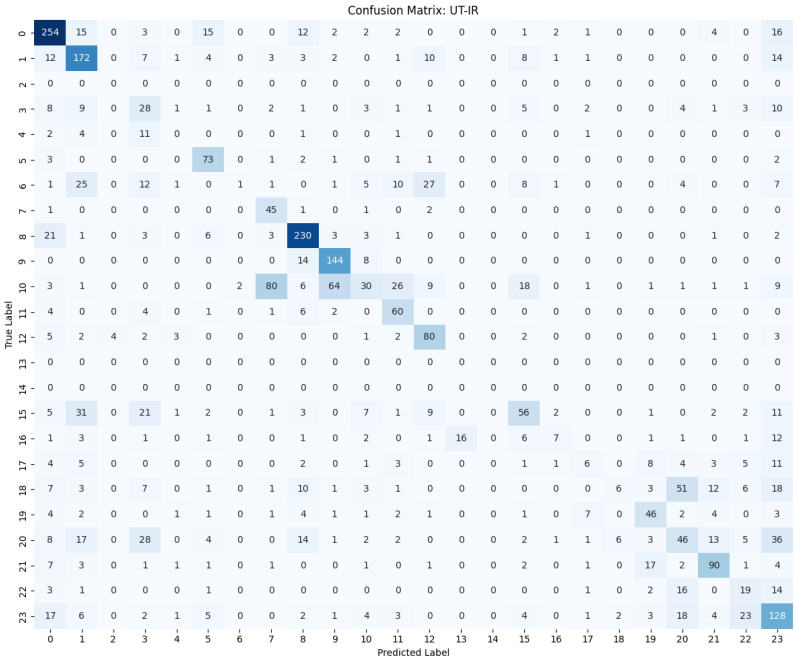


Table 3: Testing dataset (fine-tuning dataset) for Task 2. Training datasets 1 and 2 are intended for fine-tuning in downstream task.

dataset	data size	#sample in a line	# activities
SHL 2025 Task 2 testing	2860 x 500	500 (5 sec, 100 Hz)	24
SHL 2025 Task 2 training 1	3618 x 500	500 (5 sec, 100 Hz)	24
SHL 2025 Task 2 training 2	2958 x 500	500 (5 sec, 100 Hz)	29

Table 4: Table shows the provided datasets for pretrained models.

dataset for pretraining	data size	#sample in a line	#act	loc
SHL 2025 train data	196072 x 500	500 (5 sec, 100 Hz)	9	bag, hand, hips, torso
SHL 2025 valid data	28789 x 500	500 (5 sec, 100 Hz)	29	bag, hand, hips, torso
Kyutech IMU data	94997 x 500	500 (5 sec, 100 Hz)	no label	hip

a large model in combination with an elaborate fine-tuning and data augmentation pipeline. While this approach yielded strong performance, it incurred significantly higher computational costs in terms of hardware requirements, training duration, and model complexity.

TDU-BSA2. The TDU-BSA2 team based their approach on fine-tuning the MOMENT Transformer-based foundation model [5]. To mitigate overfitting and enhance robustness against the domain shift, they utilized a small, labeled subset for fine-tuning, complemented by a suite of online data augmentation techniques, including scaling, Z-axis rotation, additive white noise, and TimeWarp.

A key novelty of their method lies in the post-processing stage. The team first identified contiguous segments with constant predicted labels. A median filter was then applied within each segment to smooth the predictions. Furthermore, to precisely locate activity changes, a LightGBM classifier was trained on statistical features extracted from the raw sensor signals. This time-series-aware post-processing strategy was designed to boost performance beyond the base model's output.

SIAT-BIT. The SIAT-BIT team proposed a semi-supervised learning framework to tackle the domain shift challenge without requiring any labeled data from the target domain. Their approach is centered around a domain-aware MixMatch strategy [3]. The model, a Transformer-based classifier, was trained on handcrafted time-series features extracted using the TSFEL library.

The novelty of their method is the integration of multiple techniques within a unified training pipeline. This includes pseudo-label generation for unlabeled target data, confidence-based filtering of these labels, data augmentation, and consistency regularization. By combining supervision from multiple source domains with insights from the unlabeled target domain, the model aimed to improve its generalization capabilities in a challenging cross-domain scenario. This team was unique in its use of handcrafted features and a semi-supervised approach.

UT-IR. The UT-IR team also adopted a foundation model approach, specifically fine-tuning the large variant, MOMENT-1-large [5]. Their strategy focused on leveraging the power of large-scale pre-training combined with a sophisticated fine-tuning and data augmentation schedule to adapt the model to the target task. The team used all 15 channels of the raw IMU data, which were preprocessed into 5-second windows applying Z-score normalization and stored in LMDB format for efficient I/O. The team constructs the training set by combining two distinct training data sources and applies a stratified 80/20 split to form the training and validation sets. Then the team fine-tunes a large self-supervised foundation model (MOMENT-1-large) using on-the-fly data augmentations such as axis rotation, scaling, magnitude/time warping, phase randomization, and frequency masking. Training proceeds in three phases: an initial warm-up of 150 epochs at a 1e-3 learning rate, followed by 100 epochs at 5e-3, and a final 100 epochs at 1e-3. This is followed by 5-fold cross-validation to robustly evaluate generalization and a final fine-tuning stage on the entire dataset.

The main novelty of the team is twofold. First, the team introduced a hybrid, three-stage training regime: an initial warm-up phase with a low learning rate and conservative augmentations, a second phase with a higher learning rate for rapid convergence, and a final fine-tuning phase at a lower rate to stabilize the model. Second, the team employed a 5-fold cross-validation scheme where each fold was warm-started using the pre-trained self-supervised weights, maximizing data reuse and training efficiency. This approach represents a deep dive into optimizing the fine-tuning process of a large foundation model for a specific, challenging downstream task.

4 Analysis of Results

Overall Performance by Evaluation Metrics. Table 7 summarizes the overall performance of each team. The overall results are based on the F1-score: the first is TDU-BSA2, the second is SIAT-BIT, and the third is UT-IR.

Confusion Matrix. Figures 2 and 3 show the confusion matrices of each participants. In general, low precision indicates a high number of False Positives (FP), suggesting a tendency to misclassify instances from other classes as the target class. Conversely, low recall implies a high number of False Negatives (FN), indicating that instances of the target class are being overlooked and classified as other classes.

There are two primary categories of classes that are difficult to classify across all teams: unclassifiable classes (with extremely high FN rates) and easily misrecognized classes (with high FN rates). Unclassifiable classes (extremely high FN) include classes 2, 4, 6, 13, and 14. For these classes, the recall is close to zero across nearly all teams, meaning that almost all instances are being misclassified into other categories. Easily misrecognized classes (high FN) include classes 10, 18, 20, and 23. For these classes, all three teams exhibited recall values below 0.5, indicating frequent misclassification. Notably, class 10, despite having a large support (253 instances), showed very low recall values (TDU-BSA2: 0.1462, UT-IR: 0.1186, SIAT-BIT: 0.1265). Classes 3, 17, and 20 are prone to inducing misclassification (high FP). All three teams demonstrated low precision for these classes, suggesting that predictions for these classes often included many instances from other classes. Particularly, classes 3 and 20 showed both low precision and low recall, indicating the most challenging confusion patterns.

Each team also exhibited characteristic confusion tendencies. For TDU-BSA2, classes 10, 12, and 16 showed a distinct *high precision, low recall* pattern, suggesting a conservative confusion strategy-aiming to minimize FPs at the cost of increased FNs. On the other hand, the UT-IR demonstrated a *high recall, low precision* pattern in classes 7 and 12, suggesting a more aggressive confusion pattern-attempting to capture more true positives but inadvertently increasing FPs. It is worth noting that approximately 85% of instances of class 10 were missed. Having information on which specific classes they were misclassified as would allow for deeper insight into inter-class similarity and feature overlap, providing invaluable guidance for improving model performance.

ANOVA on Per-Class F1 Scores. We tested whether the differences in mean F1-scores across the 24 classes for each team were statistically significant using one-way ANOVA. TDU-BSA2's model exhibited the highest overall performance, and the performance gap was found to be statistically significant. The setting was the following: (1) Null Hypothesis (H_0): The mean per-class F1-scores are equal across the three teams, and (2) Alternative Hypothesis (H_1): At least one team's mean F1-score differs. Results were F-value = 3.59, p-value = 0.033. Since $p < 0.05$, we reject the null hypothesis and conclude that there are significant performance differences between the teams.

Spearman Rank Correlation. Spearman rank correlations coefficient were computed between the class-wise F1-scores[3]. The correlation results are presented in Table 6. The p-values indicate that all correlations are statistically significant. These findings suggest that the models consistently perceive the same classes as difficult or easy to classify. These results suggest that UT-IR and SIAT-IBT teams shared a more similar prediction trend than either did with TDU-BSA2.

Per-Class Performance. The following observations were made based on the class-wise precision, recall, and F1-scores: All teams struggled with classes 2, 4, 6, 13, and 14, where the number of samples was zero or very small. TDU-BSA2 outperformed others in classes such as 0, 5, 7, 8, 9, 11, 12. SIAT-BIT showed balanced performance in middle-performing classes like 15, 16, 17, 18, 21, 23. UT-IR showed higher performance in classes 1, 12, and 19 compared to the other two.

Qualitative Cluster Analysis. We observed the following grouping trends among classes based on F1-score behavior: (A) consistently high performance: classes 0, 1, 5, 8, 9, 11, (B) inconsistent or moderate performance: classes 10, 12, 15, 18, 21, 23, and (C) low or undefined performance: classes 2, 4, 6, 13, 14. This indicates a natural clustering in the confusion patterns: clearly classifiable, ambiguous, and poorly defined classes.

[3]Despite the absence of probability outputs, Spearman correlation remains a valuable metric for comparing model behavior patterns, particularly in identifying shared trends in class difficulty.

Table 5: Overall Results on the Challenge Testset. This time, the overall ranking was calculated in terms of F1-score.

Team	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
TDU-BSA2	0.5650	0.4976	0.4603	0.4389 (1st)
SIAT-BIT	0.5108	0.3924	0.4276	0.3921 (2nd)
UT-IR	0.5318	0.4125	0.4354	0.3902 (3rd)

Table 6: Spearman Rank Correlation of F1-Score Rankings Between Teams

Team Pair	Spearman's ρ	p-value	Correlation Strength
TDU-BSA2 vs. UT-IR	0.689	< 0.001	Strong Positive
TDU-BSA2 vs. SIAT-BIT	0.662	< 0.001	Strong Positive
UT-IR vs. SIAT-BIT	0.803	< 0.001	Very Strong Positive

5 Background Studies of MOMENT and Chronos

This section is a background study of two foundation models: Chronos and MOMENT.

Input Specification. Chronos assumes min-max normalization scaling to the range [0, 1] as a prerequisite preprocessing step for input signals. The official implementation includes this normalization through the NormalizeTransform class, which independently scales each sensor channel to the [0, 1] range. Regarding tokenization, Chronos employs an architecture analogous to a Vector Quantized Variational Autoencoder (VQ-VAE), converting continuous representations into discrete tokens. The input signals are first passed through an encoder-based on either convolutional neural networks (CNNs) or transformers-which projects them into low-dimensional feature representations. These are then quantized using a codebook, resulting in a tokenized sequence. Chronos also accounts for variable-length time series by applying techniques such as linear interpolation and padding to unify sequence lengths.

MOMENT adopts z-score normalization, which suggests zero mean, unit variance, as its default preprocessing method. This approach is widely used in benchmark datasets such as the UCR/UEA time-series archives, and the implementation standardizes each sensor channel independently. Unlike Chronos, MOMENT does not employ tokenization. Instead, it operates directly on continuous vector representations. By applying a contrastive loss, the model learns relationships among latent vectors derived from different temporal segments. For downstream tasks, the output of the encoder, which is either the full sequence of embeddings or a summarized representation, is directly used as input to classifiers or regressors. MOMENT thus assumes that the input data and learned features remain in a continuous space throughout both pretraining and task-specific fine-tuning.

Sliding Window Sizes. A mismatch in sliding window sizes between the pretraining phase and downstream tasks can be a critical issue. Chronos and MOMENT adopt different strategies to mitigate this problem.

Chronos addresses temporal scale discrepancies arising from varying window sizes by performing multi-scale pretraining with multiple window lengths (e.g., 32, 64, 128). This allows the model to learn representations that capture patterns across diverse temporal resolutions. Additionally, Chronos employs relative positional encoding within its Transformer architecture, which helps maintain performance regardless of sequence length. In the downstream phase, Chronos uses lightweight fine-tuning via head-tuning, where only the output layer is retrained while the feature extractor remains fixed. This enables flexible adaptation to different window sizes without extensive retraining.

MOMENT targets multimodal time-series data that span varying temporal scales. It utilizes cross-scale contrastive learning, training the model to produce semantically consistent representations across different window lengths. To further enhance robustness to temporal scale variation, MOMENT introduces consistency regularization. For feature integration, MOMENT replaces fixed-length pooling with an attention-based fusion module, which enables stable representation learning even when input sequence lengths vary significantly.

Sampling Frequency. Discrepancies in data sampling frequency between the pretraining and downstream stages can lead to significant performance degradation. Both models implement distinct strategies to mitigate the adverse effects of such inconsistencies.

Chronos adopts an architecture that explicitly captures multi-scale temporal features. It employs temporal encoding that incorporates periodic patterns such as time-of-day and day-of-week, enabling the model to learn semantically consistent representations across different temporal resolutions. Furthermore, it incorporates a multi-scale encoder that decomposes the input time series into various temporal scales and processes each scale in parallel. This design allows the model to effectively capture both high-frequency and low-frequency components. As a result, even when the sampling frequency differs between the pretraining and downstream phases, the model can transfer cross-scale representations in a robust manner.

MOMENT is built upon the Mamba architecture, which inherently supports long-range dependencies through a novel sequence modeling paradigm that is neither autoregressive nor attention-based. This structure offers flexible handling of temporal interval

Table 7: As part of the background study, the organizers conducted experiments using the Chronos and MOMENT foundation models on the testset of challenge datasets. In particular, the pretraining of the Chronos* variant was performed without the multi-scale strategy that leverages multiple window sizes. The results, illustrated in this Table, demonstrate that this omission significantly impacted performance: in five out of six runs, performance decreased, with only one run showing improvement. In the table, "SHL2025" refers to the four subsets of the SHL2025 dataset, while "Capture24" denotes the data collected from participants P001 to P150. MOMENT LR denotes logistic regression while MOMENT RF denotes random forest after obtaining the representation by MOMENT.

method	pretrained model	pretraining data	fine-tuning	features	Acc	Prec	Rec	F1-score
MOMENT	MOMENT-1-base	none	train1/2	all	0.5563	0.3999	0.4185	0.3812
MOMENT	MOMENT-1-base	none	train1	all	0.5531	<u>0.4641</u>	<u>0.4646</u>	0.4350
MOMENT	MOMENT-1-base	none	train1/2	accel	0.5133	0.3692	0.3971	0.3668
MOMENT SVM	MOMENT-1-base	none	none	all	0.3353	0.2750	0.2230	0.2008
MOMENT LR	MOMENT-1-base	none	none	all	0.3612	0.3645	0.2573	0.2414
MOMENT RF	MOMENT-1-base	none	none	all	0.5080	0.4182	0.3813	0.3781
chronos	t5-large	none	none	all	0.3923	0.3143	0.3107	0.2950
chronos	chronos_bolt_base	none	train1/2	accel	0.4829	0.3609	0.3654	0.3431
chronos	chronos_bolt_base	none	train1/2	all	0.5441	0.4411	0.4472	0.4307
chronos	chronos_bolt_base	none	train1	all	<u>0.5580</u>	0.4526	0.4539	<u>0.4362</u>
chronos	chronos_bolt_small	none	train1/2	all	0.5469	0.4406	0.4518	0.4258
chronos	chronos_bolt_mini	none	train1/2	all	0.5517	0.4088	0.4059	0.3880
chronos	chronos_bolt_tiny	none	train1/2	all	0.5392	0.4038	0.4051	0.3862
chronos	t5-small	none	train1/2	all	0.5248	0.4211	0.4238	0.3996
chronos*	t5-small	chronos+capture24+SHL2025	train1/2	all	0.5161	0.4233	0.4320	0.4044
chronos*	t5-small	chronos+capture24	train1/2	all	0.5217	0.4128	0.4111	0.3900
chronos*	t5-small	chronos+SHL2025	train1/2	all	0.5441	0.4169	0.4398	0.4095
chronos*	t5-small	capture24+SHL2025	train1/2	all	0.4871	0.3635	0.3881	0.3615
chronos*	t5-small	capture24	train1/2	all	0.4392	0.2927	0.3276	0.2944
chronos*	t5-small	SHL2025	train1/2	all	0.4745	0.3550	0.3650	0.3444

variations. MOMENT tokenizes the input by dividing the time series into fixed-length patches, upon which pretraining is conducted. This patching strategy localizes the effects of sampling frequency variations within each patch, thereby encouraging the model to learn more generalizable temporal structures. By focusing learning at the patch level, the model can construct representations that are less sensitive to the specific temporal resolution of the input sequence.

Class Imbalance. In Chronos, the pretraining phase employs self-supervised learning, which inherently mitigates the direct impact of label imbalance. However, when certain activities or states appear with disproportionate frequency in the data, the learned representations can become biased. To address this, Chronos incorporates temporal masking and random perturbations at the variable level during token-level time-series prediction. These mechanisms are designed to prevent the model from overfitting to frequent patterns and encourage the acquisition of diverse temporal dynamics. In the downstream stage, where class imbalance directly affects performance, Chronos typically employs strategies such as class-weighted loss functions and oversampling of minority classes. Furthermore, representations obtained through self-supervised pretraining are expected to remain robust even in the presence of skewed label distributions, often enhancing performance on underrepresented classes.

MOMENT takes a further step by emphasizing the learning of class-agnostic representations through multiple self-supervised objectives. These include tasks such as reconstructing inter-sensor relationships and preserving temporal order consistency, which do not rely on labels but enforce structural learning. By incorporating such objectives, MOMENT facilitates the learning of representations that are less susceptible to label distribution biases. In downstream scenarios, MOMENT also benefits from ensemble strategies and balanced classifiers tailored to the specific task, thereby improving accuracy on minority classes.

Distributional Discrepancies. Distribution shift (or covariate shift) can be problematic in time-series data, which are often characterized by strong non-stationarity. Chronos is designed as a general-purpose model capable of handling a wide range of time-series data through a unified representational framework. During pretraining, it is exposed to a variety of data distributions, equipping it with the robustness needed to handle distribution shifts stemming from differences in data collection conditions. Notably, Chronos includes layers that flexibly integrate tokenized representations of individual variables, enabling it to adapt to new sensors or variables without requiring structural changes. Furthermore, Chronos is built with fine-tuning in mind, allowing pretrained representations to be smoothly adapted to new environments, thereby mitigating the impact of distributional discrepancies.

MOMENT explicitly captures the inter-variable dependencies in time-series data as graph structures, enhancing its adaptability to changes in sensor configurations or data acquisition methods. Through self-supervised pretraining, MOMENT learns to embed semantically similar patterns into a shared representational space, even when they originate from different distributions. This contributes to its robustness against domain shift. Additionally, MOMENT adopts a modular architecture that decouples general-purpose representations from task-specific predictors, thereby enhancing its flexibility in adapting to diverse downstream tasks.

Input Modalities. Chronos is designed to handle varying input modalities in a unified manner. It represents time-series data as a sequence of tokens, with each sensor or variable encoded as an individual token. These tokens are then processed collectively by a Transformer architecture. This design enables the model to process different sensor types or configurations through the same internal mechanisms and map them into a shared representational space. Chronos employs a tokenizer that can adjust encoding strategies based on the type of variable making it inherently suitable for multimodal tasks involving heterogeneous physical quantities.

MOMENT explicitly models sensor structures and their interrelationships as graphs, where each sensor is treated as a node. This approach captures interactions between modalities and leverages structural patterns to robustly extract information even when sensor types or configurations vary.

Data Augmentation. In Chronos, data augmentation is explicitly designed to support contrastive learning by generating diverse views of the same input signal. The model relies on the assumption that different augmentations of the same time series should yield similar representations in the latent space. To achieve this, the official implementation incorporates a suite of transformation techniques, including time cropping, time reversal, amplitude scaling, and the injection of Gaussian noise. These transformations are orchestrated through a dedicated DataTransform class, which produces two distinct yet semantically related versions of each input sequence-one serving as the query and the other as the key in the contrastive learning objective. This explicit augmentation strategy is central to Chronos's ability to learn robust and invariant representations from unlabeled time-series data.

MOMENT, which is based on Contrastive Predictive Coding, adopts a more implicit approach to data augmentation. While the model does not rely on an extensive set of explicit signal transformations, it leverages the temporal structure of time series to create learning signals. Specifically, MOMENT operates on windowed subsequences extracted from longer sequences, with the objective of predicting future latent representations from current context windows. This temporal prediction task, in itself, acts as a form of augmentation by exposing the model to varied temporal contexts. Additionally, techniques such as temporal shifting and input masking are employed to further enrich the learning signal without converting the input data into discrete tokens or applying strong artificial distortions.

6 Conclusions

This summary paper presents the outcomes of Task 2 of the Sussex-Huawei Locomotion Challenge. Out of six registered teams, three submitted valid results: TDU-BSA2, UT-IR, and SIAT-BIT.

Based on the official evaluation metric, the F1 score, the first place was awarded to TDU-BSA2 (Tokyo Denki University), followed by SIAT-BIT (Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences) in second place, and UT-IR (University of Tehran) in third place. This paper provides a comparative analysis of the methodologies employed by these three teams and presents additional experiments evaluating the performance of the MOMENT and Chronos foundation models on the fine-tuning dataset used in the competition.

Acknowledgement

This research is supported by the Japan Society for the promotion of science (Grant Numbers 24K15107). The dataset used in this work was supported by HUAWEI Technologies within the project "Activity Sensing Technologies for Mobile Users".

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